

CHAPTER 1

EXECUTIVE SUMMARY

1.1 Overview of the Business Idea

This business idea aims to an innovative platform designed to tackle the growing problem of mobile e- waste by simplifying the recycling and reuse of old mobile phones and their scrap components. With the rapid growth in smartphone adoption, millions of devices are discarded annually, contributing to a significant environmental challenge. Eco Mobile aims to address this issue by providing a comprehensive and sustainable solution that benefits users, the environment, and the circular economy. At its core, Eco Mobile offers a user-friendly app that allows individuals to easily dispose of their old mobile devices or scrap components responsibly. Through a seamless interface, users can list items for recycling or resale, ensuring these materials are reintegrated into the supply chain rather than ending up in landfills. The platform also provides an accurate price estimation tool to determine fair market values for phones and scrap parts, ensuring transparency and trust in transactions.

Eco Mobile encourages reuse by creating a marketplace for buying affordable, used phone components. This not only extends the lifespan of electronic products but also offers a cost-effective solution for those who need parts for repairs or upgrades. By promoting reuse, the platform contributes to reducing the demand for new raw materials, thus lowering environmental impact. To educate users about the importance of proper e-waste management, Eco Mobile includes a resource hub offering information on the environmental and health risks associated with improper disposal. The platform provides guidance on recycling best practices, fostering a sense of environmental responsibility among its users. The business model ensures smooth transactions by integrating secure payment methods and efficient logistics support for collecting and delivering devices. Partnerships with recycling firms and refurbishing companies further strengthen the ecosystem, enabling a sustainable and scalable operation.

Eco Mobile's vision extends beyond being a platform—it aims to create a movement that promotes responsible consumption and waste management in the electronics sector. By addressing key customer pain points, such as the inconvenience of recycling and a lack of awareness, Eco Mobile stands out as a pioneer in mobile e-waste management.

1.2 Vision & Mission

- **Vision:** To create a sustainable future by reducing mobile e-waste and promoting the reuse and recycling of mobile components, fostering a circular economy.
- **Mission:** To provide an accessible, user-friendly platform that facilitates the responsible disposal, recycling, and reuse of mobile e-waste, while raising awareness and ensuring fair market practices for all stakeholders.

1.3 Key Highlights

- **Target Market:**
The target market includes eco-conscious individuals, mobile users, repair shops, and Businesses seeking sustainable e-waste solutions.
- **Unique Selling Proposition (USP):**
Eco Mobile offers a seamless platform for responsible mobile e-waste disposal recycling and reuse, ensuring fair pricing and environmental sustainability.

1.4 LOGO



Fig No: 1.1 Logo of the venture

CHAPTER 2

THE PROBLEM AND OPPORTUNITY

2.1 Problem Statement

The growing volume of mobile e-waste poses significant environmental challenges, with improper disposal and limited recycling options contributing to pollution and resource wastage. There is a lack of accessible, transparent solutions for individuals and businesses to responsibly dispose of or reuse mobile devices and components, hindering sustainable practices in the industry.

2.2 Market Opportunity

The mobile e-waste market presents a significant opportunity, as the rapid pace of technological advancements leads to an increasing number of discarded devices. With growing environmental awareness and stricter regulations on e-waste management, there is a rising demand for sustainable disposal and recycling solutions. The lack of comprehensive platforms that facilitate easy, transparent recycling and reuse creates a gap that Eco Mobile can capitalize on by offering an eco-friendly, convenient solution for individuals, repair shops, and businesses alike.

2.3 Problem Interviews and Survey Results

- 68% of respondents were unaware of proper e-waste disposal methods, indicating a need for education on recycling.
- 55% struggled to find convenient platforms for recycling or selling old devices, highlighting a market gap.
- 72% expressed interest in affordable, second-hand parts, showing strong demand for a market place focused on sustainability.

CHAPTER 3

JUSTIFICATION FOR SDG GOAL

3.1 Chosen SDG and Targets

- Primary Goal: SDG 12 – Responsible Consumption and Production
- Secondary Goal: SDG 13 –Climate Action
- Tertiary Goal: SDG 8 – Decent work and economic growth

3.2 Alignment with SDG Targets

The project fosters collaboration among key stakeholders, reduces energy wastage, and modernizes energy infrastructure, directly supporting these SDGs.

3.3 Justification and Impact

Promotes efficient energy use, ensuring affordability and sustainability.

Encourages innovation in energy management systems.

3.1 Designation & Organization:

SDG	GOAL NUMBER	GOAL NAME	TARGET NUMBER	TARGET
PRIMARY GOAL	12	Responsible Consumption and Production	12.3	Reduce 10000 mobile devices within first year.
SECONDARY GOAL	13	Climate Action	7.9	Educate 1 million individuals on e-waste management.
TERTIARY GOAL	8	Decent work and economic growth	8.5	Facilitate 20000 transactions of used mobile components.

Table 3.1 . SDG GOALS

CHAPTER 4

LITERATURE SURVEY

1. Project Title: E-Safe: An E-waste Management and Awareness Application using YOLO

Object Detection

Authors:

Khalid Alfatmi; Falguni Shashikant Shinde; Makarand Shahade; Shalaka Sanjay Sharma;
Sejal Sanjay Aruja; Tanmay Yograj Chaudhari

Relevance:

EcoMobile, aligns with this solution by addressing e-waste management through actionable features that go beyond identification. While the YOLO-based app focuses on identifying electronic devices and listing components for recycling, EcoMobile provides a comprehensive platform for easy disposal, reuse through a marketplace, fair pricing, and raising awareness about e-waste impacts. By combining user-friendly tools with educational resources, your project not only simplifies the recycling process but also empowers users to actively contribute to sustainable e-waste management, bridging the gap between technological innovation and practical implementation.

2. Project Title : Electronic waste: A critical assessment on the unimaginable growing pollutant, legislations and environmental impacts

Detection Authors:

Ramachandran Rajesh, Dharmaraj Kanakadhurga, Natarajan Prabakaran

Relevance:

EcoMobile, aligns closely with this review by turning insights on e-waste recycling and awareness programs into actionable solutions. While the review highlights the challenges and impacts of e-waste in India, along with the importance of legal measures and education, EcoMobile builds on these ideas by offering a user-friendly platform for disposal, reuse of components, fair pricing, and awareness tools. By addressing both consumer and environmental needs, your project provides a practical and sustainable approach to managing the growing e-waste crisis in high-consumption regions like India.

3. Project Title : Threats of Internet-of-Thing on Environmental Sustainability by

E-Waste Detection Authors:

Batoul Modarress Fathi 1 , Alexander Ansari 2 and Al Ansari 3

Relevance:

Eco Mobile, addresses the critical issues highlighted in this study by providing actionable solutions to mitigate the environmental risks of growing e-waste. While the study identifies Internet penetration and IoT proliferation as significant contributors to the e-waste crisis, EcoMobile offers a practical response through easy disposal options, a marketplace for reuse, fair pricing, and educational tools. By engaging users, developers, and stakeholders, your project bridges the gap between awareness and action, promoting sustainable practices to counter the rising challenges of e-waste in the digital age.

4. Project Title: Forecasting of Scrapped (e-waste) Mobiles

Authors:

Lotlikar Swapnali Vivek; Ajay Khuteta; Seema Shaha

Relevance:

EcoMobile, addresses the concerns highlighted in this study by providing a practical platform to manage the exponential growth of mobile waste. While the study estimates the patterns of mobile recycling and scrapping, EcoMobile offers solutions like easy disposal, reuse options, and educational tools to increase recycling rates and reduce waste, promoting sustainable mobile e-waste management.

5. Project Title: Reducing electronic waste through the development of an adaptable mobile device

Detecting Author:

Madeleine Brannon; Phillip Graeter; Donald Schwartz; Joost R. Santos

Relevance:

Eco Mobile, directly addresses the issues raised in this study by offering practical tools for e-waste disposal, reuse, and awareness. While the study highlights the role of IoT and Internet proliferation in increasing e-waste, your app provides actionable solutions to mitigate these effects, promoting sustainable practices and reducing environmental risks.

6. Project Title: SAFA-E (The E-Waste Management System)

Detecting Author: Vishwash Chaturvedi; Ritika Babbar; Ishita Arora; Divakar Varshney; U. Hariharan

Relevance:

EcoMobile, complements the proposed Safa-e system by offering additional features such as price estimation, a reuse marketplace, and awareness tools, enhancing e-waste disposal and recycling efforts. While both focus on proper waste management, EcoMobile empowers users with a seamless platform to actively contribute to sustainable practices and reduce environmental risks.

7. Project Title : Reduce, Reuse, Recycle (3Rs) Awareness App: Mobile Learning Application for Supporting Environmental Awareness Initiatives

Detecting Author:

Maslinda Mohd Nadzir; Nur Faisya Arnisa Sunny Seowfuddin

Relevance:

EcoMobile, aligns with this study by promoting environmental awareness through practical solutions. While the 3Rs mobile application focuses on educating children, EcoMobile targets a broader audience by providing tools for e-waste disposal, reuse, and recycling, fostering sustainable practices and empowering users to actively contribute to environmental care.

8. Project Title : Comparison of Different Computer Vision Approaches for E-waste Components Detection to Automate E-wasteDisassby

Detecting Author:

Abdelrhman; M.Bassiouny; Abdelrahman S.Farhan; Shady A. Maged; Mohammed I. Awaad

Relevance:

EcoMobile, complements this study by addressing the e-waste crisis with a user-centric platform for disposal, reuse, and recycling education. While the study focuses on automation and robotics for efficient component separation, EcoMobile empowers individuals to actively participate in sustainable e-waste management, bridging the gap between advanced technological solutions and everyday user actions.

9. Project Title : Building reverse supply chain of waste mobile phones based on e-commerce platform

Detecting Author: Menglia

Relevance:

EcoMobile, aligns with this study by leveraging a digital platform to streamline the reverse supply chain of mobile e-waste. While the paper highlights the cost and efficiency advantages of e-commerce- based recovery, EcoMobile enhances this approach by integrating easy disposal, reuse options, price estimation, and awareness tools, making mobile e-waste management more accessible and user- friendly.

10. Project Title : Reduce, Reuse, Recycle (3Rs) Awareness App: Mobile Learning Application for Supporting Environmental Awareness Initiatives

Author:

Maslinda Mohd Nadzir; Nur Faisya Arnisa Sunny Seowfuddin

Relevance:

EcoMobile, aligns with this study by promoting environmental awareness through practical solutions. While the 3Rs mobile application focuses on educating children, EcoMobile targets a broader audience by providing tools for e-waste disposal, reuse, and recycling, fostering sustainable practices and empowering users to actively contribute to environmental care.

11. Project Title : Comparison of Different Computer Vision Approaches for E-waste Components

Detection to Automate E-waste Disassembly

Author:

Abdelrhman; M.Bassiouny; Abdelrahman S.Farhan; Shady A. Maged; Mohammed I. Awaad

Relevance:

Eco Mobile, complements this study by addressing the e-waste crisis with a user-centric platform for disposal, reuse, and recycling education. While the study focuses on automation and robotics for efficient component separation, Eco Mobile empowers individuals to actively participate in sustainable e-waste management, bridging the gap between advanced technological solutions and everyday user actions.

12. Project Title : Building reverse supply chain of waste mobile phones based on e-commerce platform

Author: Menglia

Relevance:

Eco Mobile, aligns with this study by leveraging a digital platform to streamline the reverse supply chain of mobile e-waste. While the paper highlights the cost and efficiency advantages of e-commerce-based recovery, Eco Mobile enhances this approach by integrating easy disposal, reuse options, price estimation, and awareness tools, making mobile e-waste management more accessible and user-friendly.

CHAPTER 5

MARKET RESEARCH AND STRATEGY

5.1 Market Size Estimation:

The global e-waste management market was valued at approximately \$49.8 billion in 2022 and is expected to grow at a compound annual growth rate (CAGR) of 23.5% from 2023 to 2030. By focusing specifically on mobile e-waste and scrap management, Eco Mobile can target a niche within this larger market, estimated to range between \$5 to \$10 billion. This includes eco-conscious individuals, mobile repair shops, and businesses in regions with high mobile device turnover, such as North America, Europe, and Asia, offering a significant growth opportunity in the sustainable e-waste sector.

5.2 Audience and Customer Persona

- **Primary Target Audience:** The primary target audience includes eco-conscious mobile device users, repair shops, and businesses with 1-50 employees seeking sustainable solutions for e-waste disposal and reuse.

- **Customer Persona:**

Age: 25

Occupation: Software Developer

Location: Urban area with access to recycling facilities.

Income Level: Middle-class, with disposable income for sustainable purchases.

5.3 Industry Trends and Insights

Global e-waste is projected to grow by 21%, reaching 74 million metric tons by 2030, underscoring the need for better recycling solutions. 32% of consumers are willing to pay more for eco-friendly products, indicating strong demand for sustainable mobile e-waste management.

5.4 Competition Analysis

Benefits	Competitor 1	Competitor 2	Competitor 3	Competitor 4	Your Venture
Product	EcoATM	Decluttr	SellCell	Best Buy Trade -In	Mobile E-Waste and scrap Management
Price	1600	3200	4500	1600	1600
Branding channels	In-store kiosks, digital advertising.	Online ads, social media, influencer marketing.	SEO-focused website, tech blogs.	In-store promotions, email marketing.	Online advertising, affiliate marketing.
Packaging	Physical kiosks; no packaging needed.	User-friendly online interface; minimal physical packaging.	Digital service; no physical packaging required.	No packaging; in-store process.	User Friendly digital App Interfaces.
Market reviews	Generally positive for convenience; some complaints about low offers.	Positive for user experience; mixed on valuation accuracy.	Positive for transparency; users appreciate comparison features.	Mixed reviews; convenience praised but valuation complaints	Generally positive; users like the variety of offers.
UVP	Quick and convenient cash for recycled phones at accessible kiosks.	Easy online selling with fast payment for devices.	Finds the best resale value across various platforms quickly.	Convenient in-store trade-ins for easy upgrades and immediate store credit.	Access to numerous buyer offers, allowing for competitive pricing.

Fig No: 5.4 Competition Analysis

5.5 Our Advantage:

Manufacturer Programs: Apple and Samsung focus on trade-ins, not scrap parts or component resale. E-Waste Startups: Platforms like Gazelle buy phones but don't address broader mobile scrap management.

Local Centers: Recycling centers offer limited services and lack a user-friendly digital platform.

5.6 Unique Selling Proposition (USP)

Eco Mobile offers a unique platform for mobile e-waste by providing a seamless marketplace for recycling, reusing, and selling mobile scrap components. It combines ease of disposal with fair pricing, educational resources on sustainable practices, and secure transactions, addressing the growing need for eco-friendly mobile waste management while empowering users to make a positive environmental impact.

CHAPTER 6

SOLUTION AND PRODUCT DEVELOPMENT

6.1 Proposed Solution

Eco Mobile provides a comprehensive platform for managing mobile e-waste, offering users a simple way to recycle, sell, and buy used mobile components while promoting sustainability.

- **Easy Disposal:** A user-friendly interface for selling or recycling old phones and components.
- **Encourage Reuse: Marketplace** for affordable, used phone parts, reducing e-waste.
- **Raise Awareness:** Educational tools on the environmental impact of e-waste and recycling practices.

6.2 Product Features and Benefits

- **User-Friendly Interface:** Simplifies selling, recycling, and buying components. Benefit: Makes e-waste management accessible.
- **Marketplace for Reused Parts:** Facilitates buying and selling refurbished components.
- **Benefit:** Reduces e-waste and lowers repair costs.
- **Educational Resources:** Provides content on e-waste impact and recycling.
- **Benefit:** Promotes responsible disposal and reuse.

6.3 Value Proposition Canvas

- **Customer Jobs:** Easy disposal of phones and affordable repair options.
- **Pain Points:** Limited recycling options, high costs, lack of awareness.

6.4 Minimum Viable Product (MVP)

- User Registration
- Phone Recycling Process
- Marketplace for Components
- Secure Transactions

CHAPTER 7

SYSTEM DESIGN / PROJECT METHODOLOGY

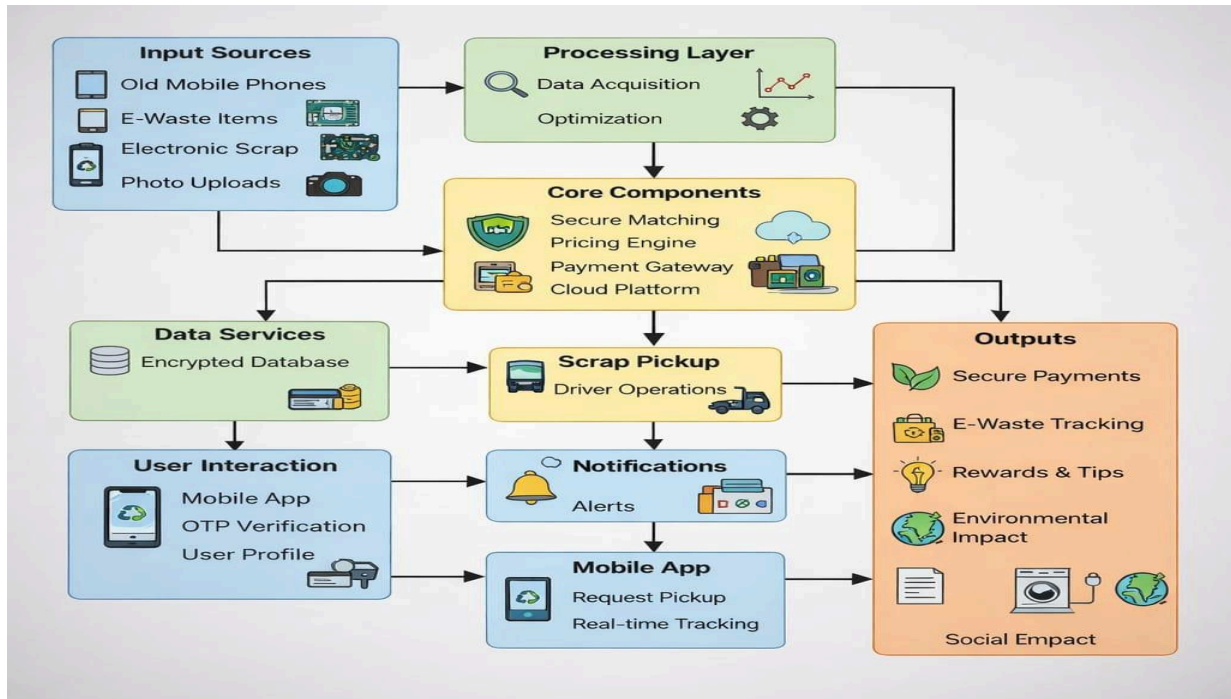


Fig 7.1. Architecture Diagram 1

Mobile E-Waste and Scrap Management App. Users act as input sources by submitting details of old mobile phones, e-waste items, electronic scrap, and photos through the mobile app. The processing layer collects and optimizes this data, which is handled by core components such as secure matching, pricing engine, payment gateway, and cloud platform. Data services store information securely in encrypted databases. The system schedules scrap pickup through driver operations and sends notifications to users. Final outputs include secure payments, e-waste tracking, rewards, and measurable environmental and social impact. The Core Components form the backbone of the system, including secure matching between users and scrap collectors, an intelligent pricing engine to estimate scrap value, a secure payment gateway, and a cloud platform for scalable operations. All user and transaction data is stored safely in Data Services using encrypted databases to maintain privacy and security. Finally, the Outputs highlight the system's impact, including secure payments, e-waste tracking, rewards and tips, environmental sustainability, and social impact, promoting responsible e-waste management and recycling.

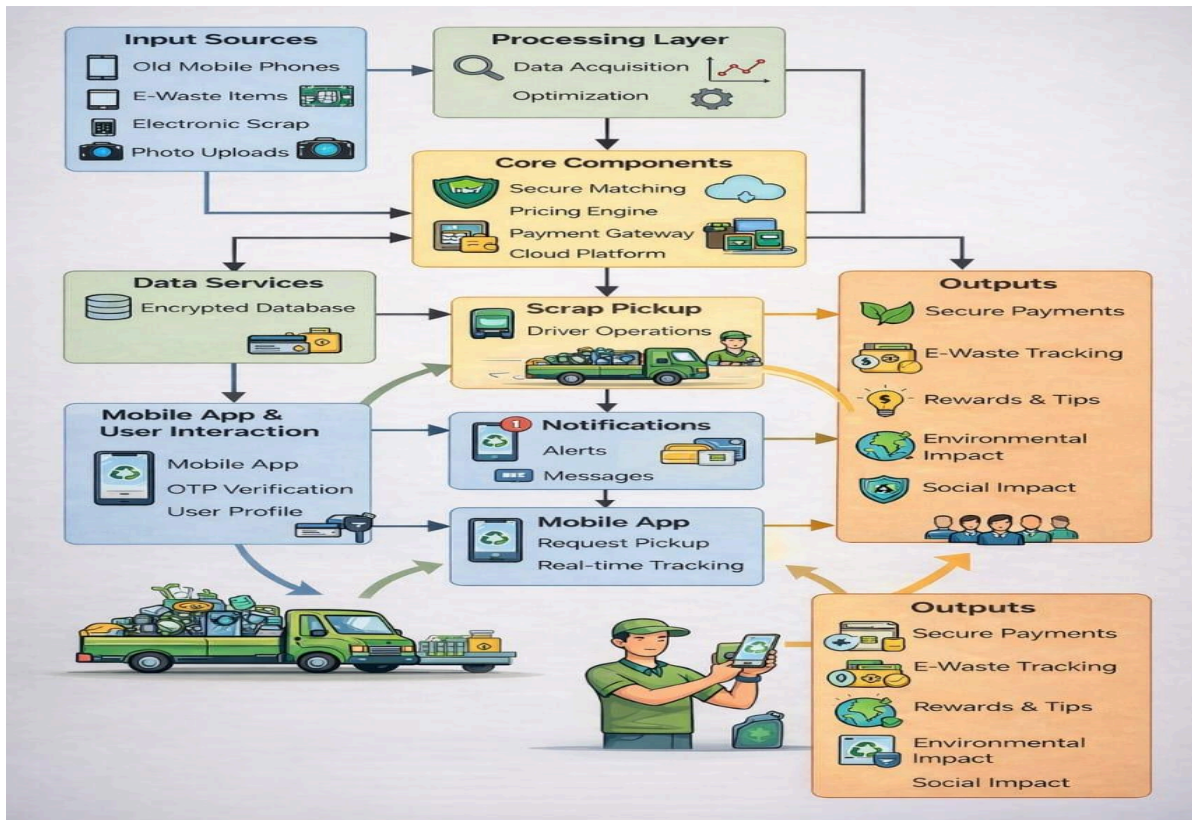


Fig 7.2 . Architecture Diagram 2

This figure illustrates an alternative architectural model of the Eco Mobile – Mobile E-Waste and Scrap Management App, emphasizing the end-to-end digital workflow of e-waste collection and recycling. The architecture begins with the Mobile App Interface, where users register, verify their identity using OTP, and submit scrap pickup requests by providing item details and images. These inputs are transmitted to the Backend Processing Layer, which validates data, categorizes e-waste types, and calculates dynamic pricing based on market value and item condition. The Core System Modules manage secure user–collector matching, transaction handling, and cloud-based storage. A centralized Database Layer maintains encrypted records of users, scrap items, payments, and pickup history. The Logistics and Pickup Module coordinates with drivers, assigns pickup schedules, and tracks real-time location updates. A Notification System ensures continuous communication by sending alerts related to pickup confirmation, driver arrival, and payment status. The system concludes with the Output Layer, which delivers secure digital payments, transparent e-waste tracking, reward incentives, and sustainability reports.

CHAPTER 8

IMPLEMENTATION

Module split ups

1. User Registration & Authentication Module
2. E-Waste Submission & Image Upload Module
3. Scrap Evaluation & Pricing Engine Module
4. Collector Matching & Pickup Scheduling Module
5. Payment Processing & Reward Management Module
6. Data Visualization, Tracking & Sustainability Impact Module
7. Notification & Communication Module
8. Data Management, Security & Admin Control Module

8.1 User Registration & Authentication Module

The User Registration & Authentication Module forms the entry point of the SCRAPZAP – Mobile E-Waste and Scrap Management System. This module enables users to securely create accounts using their mobile number or email ID, followed by OTP-based verification to ensure authenticity. It also supports login, logout, password recovery, and profile management functionalities.

User profiles store essential details such as contact information, pickup addresses, and transaction history. Secure authentication mechanisms prevent unauthorized access and protect sensitive user data. This module establishes trust and ensures that only verified users can submit scrap requests or receive payments through the platform.

8.2 E-Waste Submission & Image Upload Module

This module allows users to submit details of old mobile phones, electronic waste, and scrap items for recycling or resale.

Users can enter item type, condition, quantity, and upload images to support accurate evaluation. Image uploads help the system and collectors assess scrap quality before pickup. The module validates inputs to eliminate incomplete or false entries. Once submitted, the data is forwarded to the backend for processing. This module simplifies scrap listing and ensures transparency between users and collectors.

8.3 Scrap Evaluation & Pricing Engine Module

The Scrap Evaluation & Pricing Engine Module is responsible for estimating the value of submitted e-waste. It analyzes item category, condition, weight, and prevailing market rates to generate fair and dynamic pricing. The pricing logic ensures consistency and avoids manual negotiation. This module helps users understand the value of their scrap beforehand, increasing trust and satisfaction. It also enables collectors to make informed pickup decisions, creating a balanced ecosystem.

8.4 Collector Matching & Pickup Scheduling Module

This module manages the complete logistics and operational workflow of the system. It efficiently matches user scrap pickup requests with nearby authorized scrap collectors using location-based matching techniques, ensuring faster response times and reduced transportation distance. Pickup schedules are automatically generated by considering collector availability, distance, and time slots, enabling optimized route planning. The module provides real-time pickup status updates, allowing users to track the collector's movement and view the estimated arrival time. By coordinating logistics in an organized and systematic manner, this module ensures timely, efficient, and reliable scrap collection while improving overall service quality and user satisfaction.

8.5 Payment Processing & Reward Management Module

After the successful pickup and verification of scrap materials, this module is responsible for handling secure digital payment transactions through UPI or wallet-based payment methods. The system ensures encrypted and reliable financial processing to maintain user data security and transaction integrity. Once the payment is successfully completed, the confirmation details, including transaction status and amount credited, are automatically updated in the user's account in real time. This module enhances transparency, builds user trust, and ensures a smooth and efficient end-to-end scrap disposal and payment experience.

8.6 Data Visualization, Tracking & Sustainability Impact Module

This module provides users with visual insights into their scrap disposal activity. Dashboards display total e-waste recycled, earnings, reward points, and environmental impact such as reduced landfill waste. Simple charts and indicators help users understand their contribution toward sustainability. This module increases awareness and reinforces eco-friendly habits.

8.7 Notification & Communication Module

The Notification Module plays a crucial role in keeping users informed at every stage of the transaction process. It sends timely alerts for important events such as pickup request confirmation, collector assignment and arrival, successful scrap pickup, payment completion, and reward or points updates. The module uses push notifications and in-app messages to ensure real-time communication with users. By providing continuous updates and clear information throughout the entire transaction lifecycle, this module improves user engagement, enhances transparency, and builds trust in the overall system.

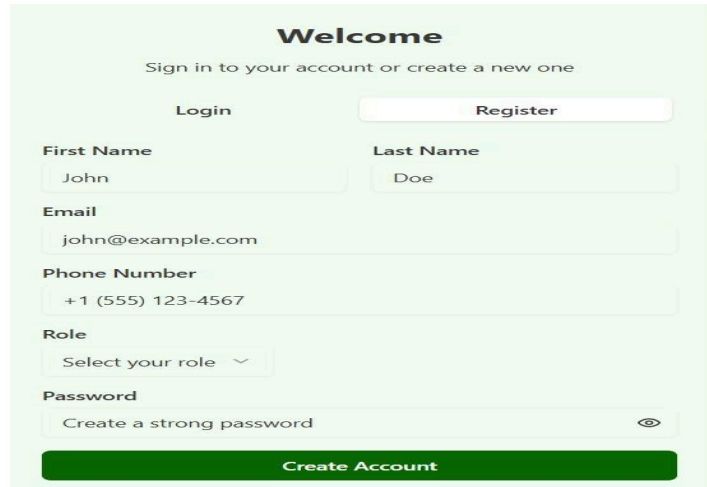
8.8 Data Management, Security & Admin Control Module

This module ensures secure storage and management of all system data. User information, transactions, and scrap records are stored in encrypted cloud databases. Role-based access control differentiates users, collectors, and administrators. Admin controls allow monitoring of requests, pricing updates, and dispute handling. Security mechanisms such as encrypted APIs, session tokens, and access validation ensure SCRAPZAP remains scalable, reliable, and secure.

CHAPTER 9

RESULTS AND ANALYSIS

Sample Output



The image shows a user registration and login interface with a light green background. At the top, it says "Welcome" and "Sign in to your account or create a new one". Below this are two buttons: "Login" and "Register". The registration form includes fields for "First Name" (John), "Last Name" (Doe), "Email" (john@example.com), "Phone Number" (+1 (555) 123-4567), "Role" (a dropdown menu with "Select your role" and a downward arrow), and "Password" (a field with the text "Create a strong password" and an eye icon). A large green button at the bottom says "Create Account".

Fig 9.1 . User Registration and Login Interface

This figure represents the User Registration and Login page of the *ScrapZap (EcoMobile)* application. This interface serves as the entry point for users to access the system securely. It allows both new users to create an account and existing users to log in, ensuring controlled and authenticated access to the platform. The registration form collects essential user details such as first name, last name, email ID, phone number, role selection, and password. The inclusion of role selection helps differentiate between various stakeholders of the system, such as users, service personnel, or administrators. This role-based approach supports better access control and system organization. A strong password creation feature is provided to enhance security and protect user data. The clean and user-friendly design ensures ease of use, even for first-time users. Overall, this interface plays a crucial role in maintaining data security, user authentication, and smooth onboarding, which are essential for the effective functioning of the ScrapZap mobile e-waste management system.

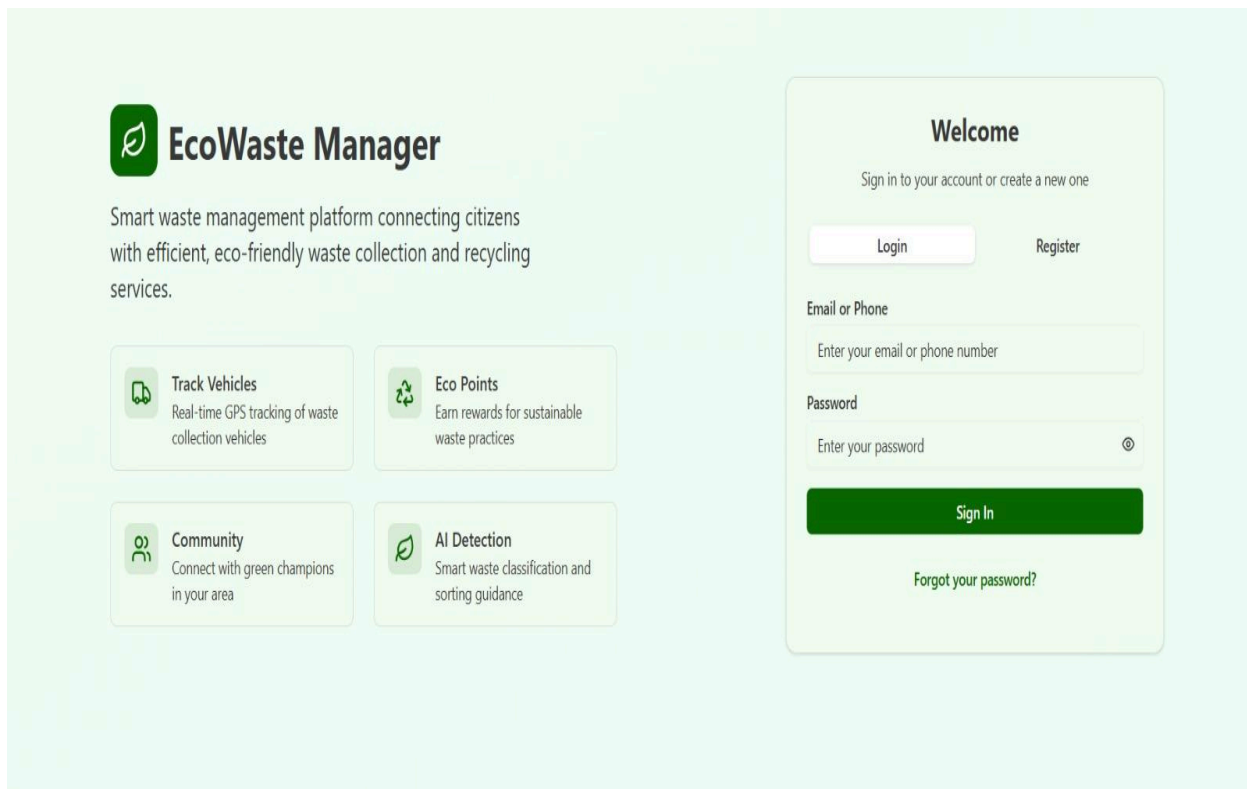


Fig 9.2 . Home Page and User Login Dashboard

This figure illustrates the Home Page and Login Dashboard of the *ScrapZap (EcoMobile)* application, presented here as the EcoWaste Manager interface. The page introduces the platform as a smart and eco-friendly waste management system that connects citizens with efficient waste collection and recycling services. It highlights the core objective of promoting sustainable practices through digital solutions. The left section of the interface showcases the key features of the system, including real-time vehicle tracking for waste collection, eco-points for rewarding responsible waste disposal, community engagement to connect users with local green initiatives, and AI-based waste detection for smart classification and sorting guidance. These features emphasize the intelligent and technology-driven approach of the application. The right section contains the secure login module, where registered users can sign in using their email or phone number and password. A “forgot password” option is provided to ensure account recovery and user convenience. Overall, this dashboard serves as an informative and functional entry point, combining feature awareness, user authentication, and sustainability goals in a single interface.

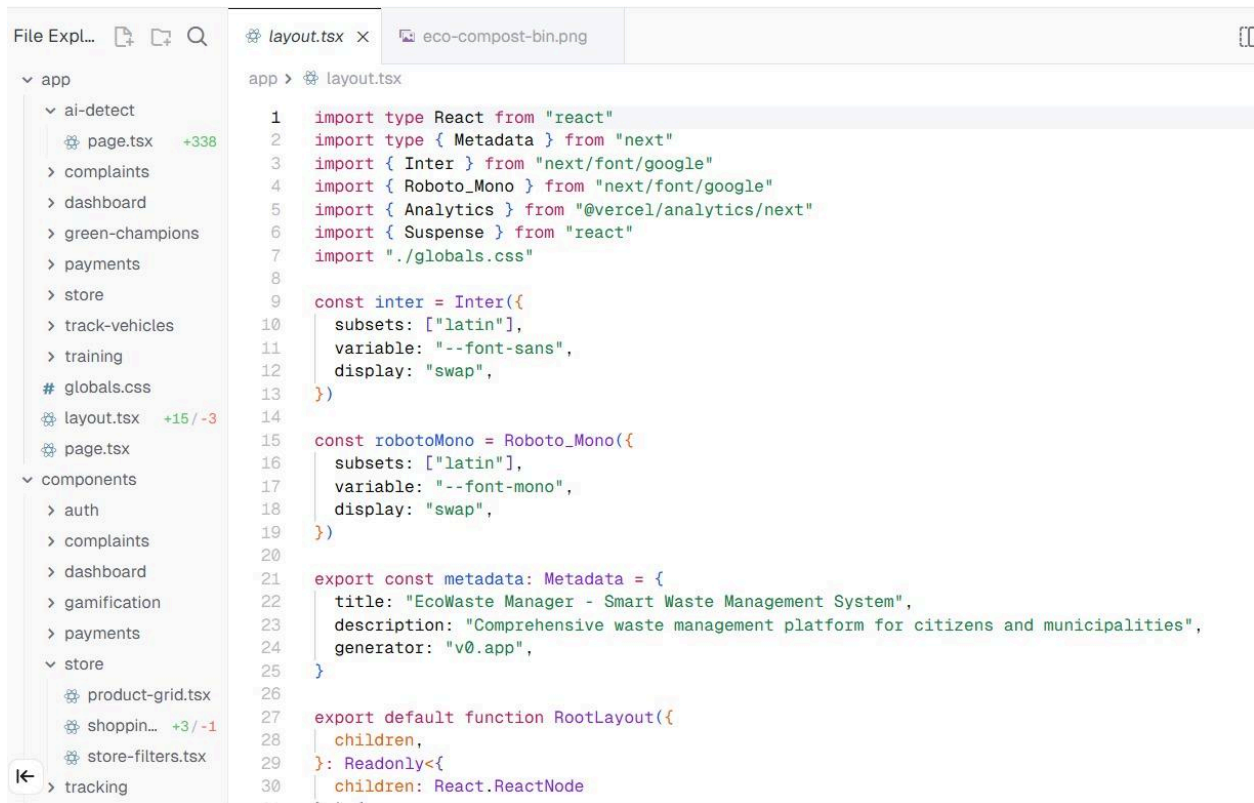


Fig 9.3 . Application Architecture and Layout Configuration

This figure shows the source code structure and layout configuration of the *ScrapZap (EcoMobile)* web application developed using Next.js and React. The file hierarchy displayed on the left illustrates a modular and scalable architecture, where each major feature such as AI detection, complaints, dashboard, payments, vehicle tracking, community engagement, and training is organized into separate folders. This structure improves maintainability and allows easy future expansion. The code displayed (**layout.tsx**) defines the global layout of the application, ensuring consistency across all pages. It includes the integration of Google fonts such as Inter and Roboto Mono, which enhance readability and provide a modern user interface. Global styling is managed through a centralized CSS file, ensuring uniform design standards throughout the application. Additionally, the layout configuration includes metadata definitions such as the application title and description, which are essential for browser display and search engine optimization. The use of reusable components, analytics integration, and structured imports reflects good software .

CHAPTER 10

BUSINESS MODEL AND STRATEGY

10.1 Business Model Overview:

- **Pro (from month 4):** Subscription for advanced features like price estimation and priority listings.
- **Enterprise (from month 6):** Custom subscriptions for bulk recycling and analytics.

10.2 Revenue Model:

Eco Mobile will generate revenue through subscription fees from Pro and Enterprise tiers, transaction fees on sales and recycling, and partnerships with mobile retailers and eco-friendly brands.

10.3 Sales plan and Go to market Strategy:

- Sales plan: use a freemium model, upsell features, and partner with retailers and Eco-Organizations
- Go-to-Market Strategy: launch with digital campaigns and local eco-events to build trust and drive user acquisition.

10.4 Marketing and Customer Acquisition Strategy:

- Targeted Digital Marketing: Use social media, Google ads, and SEO to attract eco-conscious users, focusing on e-waste education.
- Strategic Partnerships: Collaborate with NGOs, mobile brands, and local businesses.

CHAPTER 11

FINANCIAL PLAN& PROJECTIONS

11.1 Financial Overview:

- **Startup Costs:** ₹455000 (development, marketing, initial operations)
- **Revenue Projections:** ₹9140000 in Year 1.

Starting Costs	
Startup Cost	₹4,55,000
Business Registration	₹25,000
Licenses	₹10,000
Equipment and Utensils	₹2,00,000
Interiors and Furnitures	₹1,00,000
Website development	₹1,20,000
Capital Work in Progress (Fixed Asset)	₹2,00,000
Workstation	₹2,00,000
Starting Operations (Budgeted)	₹8,60,000
Promotion	₹50,000
Hosting and Domain Space	₹10,000
Miscellaneous	₹50,000
	₹7,50,000
Start-up capital	
Share Capital	₹7,50,000

Fig No: 11.1 Financial Overview

Forecasted Profit & Loss (P&L):

Year 1: Revenue: ₹108,300; Net Profit: ₹56200

Year 2: Revenue: ₹216600; Net Profit: ₹20660

Profit & Loss Account (Income Statement)													
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Totals
Sales	Gross Profit = Sales - COGS		Net Profit = Sales - COGS - Fixed Expenses										
Cash Sales	₹ 4,00,000	₹ 4,60,000	₹ 5,20,000	₹ 5,80,000	₹ 6,40,000	₹ 8,00,000	₹ 8,60,000	₹ 9,20,000	₹ 9,80,000	₹ 10,40,000	₹ 11,00,000	₹ 11,60,000	₹ 94,60,000
No. of Customers	200	230	260	290	320	400	430	460	490	520	550	580	4730
Price Per Unit	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000	₹ 2,000
Other Cash Receipts													
Total Sales	₹ 4,00,000	₹ 4,60,000	₹ 5,20,000	₹ 5,80,000	₹ 6,40,000	₹ 8,00,000	₹ 8,60,000	₹ 9,20,000	₹ 9,80,000	₹ 10,40,000	₹ 11,00,000	₹ 11,60,000	₹ 94,60,000
COGS													
Cost of Services	₹ 58,000	₹ 66,700	₹ 75,400	₹ 84,100	₹ 92,800	₹ 1,16,000	₹ 1,24,700	₹ 1,33,400	₹ 1,42,100	₹ 1,50,800	₹ 1,59,500	₹ 1,68,200	₹ 13,71,700
Cost Per Unit	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290	₹ 290
Gross profit	₹ 3,42,000	₹ 3,93,300	₹ 4,44,600	₹ 4,95,900	₹ 5,47,200	₹ 6,84,000	₹ 7,35,300	₹ 7,86,600	₹ 8,37,900	₹ 8,89,200	₹ 9,40,500	₹ 9,91,800	₹ 80,88,300
Fixed Expenses													
Salaries	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 4,25,000	₹ 51,00,000
Marketing and Promotion	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 40,000	₹ 4,80,000
Digital Marketing Utilities (Electricity etc.)	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 1,00,000	₹ 12,00,000
Customer Service	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 15,000	₹ 1,80,000
Rent	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 5,000	₹ 60,000
Total Expenses	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 80,000	₹ 9,60,000
Result	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 6,65,000	₹ 79,80,000
Net Profit/Loss	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -	₹ -
Gross Profit Margin	86%	86%	86%	86%	86%	86%	86%	86%	86%	86%	86%	86%	86%
Net Profit Margin	-81%	-89%	-42%	-29%	-18%	2%	8%	13%	18%	22%	25%	28%	1%

Fig No: 11.2 Forecasted profit and loss

Unit Economics:

Customer Acquisition Cost (CAC):

₹355 Customer Lifetime Value (CLV): ₹6471

Payback Period: 1 year

	Year 2024								
P & L/ unit			Unit Economics Year 2024			Step 1	APV = Average Purchase Value = Total revenue/Total number of purchases	₹	2,000
Revenue	₹ 2,000		CAC	₹ 355		Step 2	APF = Average Purchase Frequency = Number of purchases/ Number of unique customers		18.92
COGS	₹ 290		CLV	₹ 6,471		Step 3	Average Customer Lifespan = 1/Churnrate		0.20
Gross Profit	₹ 1,710		ARPU	₹ 37,840		Step 4	CLV = APV*APF*ACL*GM	₹	6,471
Gross Profit Margin	86%								
Operating Costs	₹ 1,977								
Operating Profit	₹ 22.90		CAC = (M + S) / Number of Customers						
			1000000	666.666667					
Assumptions			1500						

Fig No: 11.3 Unit Economics

Funding Plan and Capital Requirements:

- Product Development and Operations (40-50%)
- Marketing and Customer Acquisition (30-35%)
- Contingency and Growth Reserves (15-20%)

CHAPTER 12

TEAM & EXECUTION

12.1 Team Composition and Roles:

CEO: ASWINI PRIYA A

CTO: SAFIYA BI A

COO: HARINI T

SUPERVISOR: Mrs. S. KAVITHA

Milestone and Execution Roadmap:

- Platform Launch: Develop and release Basic tier with key features and initial marketing.
- Pro Tier & Growth: Introduce Pro features, expand user base through marketing and partnerships.
- Enterprise & Scaling: Launch Enterprise solutions, scale operations, and refine platform features.

Risk Analysis and Mitigation Strategies:

- **Risk:** Challenges include low adoption, technical issues, competition, and regulatory changes.
- **Mitigation:** Address risks through user incentives, regular testing, innovation, and compliance with regulations.

CHAPTER 13

CONCLUSION & NEXT STEPS

13.1 Summary of Key Points

Eco Mobile is a sustainable platform addressing the growing challenge of mobile e-waste by providing efficient solutions for recycling and reuse. It targets environmentally conscious users, budget-conscious buyers, and responsible businesses, offering features that integrate environmental sustainability with user convenience. The platform simplifies e-waste disposal by enabling users to sell or recycle old mobile phones and components through a user-friendly app. It incorporates price estimation tools to ensure fair valuations, fostering transparency and trust. Additionally, Eco Mobile creates a marketplace for refurbished and used phone parts, encouraging reuse and reducing the demand for new resources.

Educational resources are a core feature, raising awareness about the environmental and health hazards of improper e-waste disposal. By promoting responsible practices, Eco Mobile helps users make informed decisions that contribute to sustainability. EcoMobile aligns with multiple SDG goals, including Responsible Consumption and Production (SDG 12), by fostering recycling and reuse, and Climate Action (SDG 13), by reducing carbon footprints associated with manufacturing new devices. It also supports Decent Work and Economic Growth (SDG 8) by creating jobs in e-waste management and providing economic opportunities in the secondary electronics market.

13.2 Future Scope

The ScrapZap (EcoMobile) project has wide future scope and can be expanded into a complete digital ecosystem for electronic waste management. In the future, the application can be extended to include other types of e-waste such as laptops, batteries, chargers, and household electronics, making it a comprehensive platform. Integration of AI and Machine Learning can enable automatic scrap quality analysis and accurate price prediction using uploaded images, improving transparency and efficiency. IoT-enabled smart collection bins and GPS tracking can be introduced to optimize scrap collection and logistics. The use of blockchain technology can ensure secure, transparent, and tamper-proof

transactions while tracking the e-waste lifecycle from collection to recycling. Partnerships with government bodies, NGOs, and certified recyclers can help ensure regulatory compliance and large-scale adoption. Additionally, the reward system can be enhanced into green credits or carbon points, encouraging user participation, while multi-language support and regional scalability can make ScrapZap a globally accessible and impactful solution for sustainable e-waste management.

13.3 Reflection the Live-In-Lab I Experience

Participating in **Live-in-Lab 1** marked the true beginning of our SECMS journey. At this stage, we did not possess any hardware components, prototypes, or technical modules—only a strong idea built on extensive observation and a clear understanding of the energy consumption problem faced by households. Live-in-Lab 1 pushed us to present our concept with confidence, focusing on identifying the root cause of rising electricity usage and proposing our WattFinder-powered appliance-level monitoring solution as the answer. Even without a working model, this phase taught us how to transform a raw idea into a structured and impactful pitch. We learned to analyze user pain points, communicate our value proposition effectively, and understand the market gap. The mentor interactions during this stage became the foundation upon which our entire project steadily evolved.

13.4 Reflection of the Live-In-Lab II Experience

Live-in-Lab 2 represented the transition point where our idea began moving toward early-stage implementation. Building upon the insights, feedback, and clarity gained in Lab 1, we started working on essential feature-level components of SECMS. During this phase, we introduced the goal-tracking system, which allows users to set monthly power consumption targets and monitor progress in real time, strengthening the behavioural change aspect of our solution. We also began integrating secure third-party, government-authorized payment gateway options to ensure trust, transparency, and compliance for future bill payments. This stage helped us understand system feasibility, the logical flow of user journeys, and the importance of modular development. For the first time, our initially theoretical concept started transforming into a recognizable and functional system, giving us confidence that the complete SECMS platform was achievable.

REFERENCE

- [1]. Khalid Alfatmi and Falguni Shashikant Shinde, “E-Safe: An E-waste Management and Awareness Application Using YOLO Object Detection,” *International Journal of Engineering and Advanced Technology*, 2023.
- [2]. Ramachandran Rajesh, “Electronic Waste: A Critical Assessment on the Unimaginable Growing Pollutant, Legislations and Environmental Impacts,” *Environmental Science and Pollution Research*, 2022.
- [3]. Batoul Modarress Fathi, “Threats of Internet of Things on Environmental Sustainability by E-Waste,” *Journal of Sustainable Computing*, 2021.
- [4]. Lotlikar Swapnali Vivek, “Forecasting of Scrapped (E-Waste) Mobiles,” *International Journal of Computer Applications*, 2022.
- [5]. Madeleine Brannon, “Reducing Electronic Waste Through the Development of an Adaptable Mobile Device,” *Journal of Cleaner Production*, 2020.
- [6]. Vishwash Chaturvedi, “SAFA-E: The E-Waste Management System,” *International Conference on Sustainable and Smart Technologies*, 2021.
- [7]. Maslinda Mohd Nadzir, “Reduce, Reuse, Recycle (3Rs) Awareness App: Mobile Learning Application for Supporting Environmental Awareness Initiatives,” *Procedia Computer Science*, 2019.
- [8]. Abdelrhman and M. Bassiouny, “Comparison of Different Computer Vision Approaches for E-Waste Components Detection to Automate E-Waste Disassembly,” *IEEE Access*, 2022.
- [9]. Du Zhiping and Lang Menglian, “Building Reverse Supply Chain of Waste Mobile Phones Based on E-Commerce Platform,” *International Journal of Supply Chain Management*, 2020.

- [10]. Maslinda Mohd Nadzir, “Reduce, Reuse, Recycle (3Rs) Awareness App for Environmental Education,” *International Journal of Interactive Mobile Technologies*, 2019.
- [11]. Abdelrhman and M. Bassiouny, “Computer Vision Techniques for Automated E-Waste Component Identification,” *International Conference on Artificial Intelligence and Applications*, 2021.
- [12]. Du Zhiping, “Reverse Logistics and E-Commerce Based Recycling Model for Waste Mobile Phones,” *Journal of Environmental Management*, 2020.

APPENDIX I

SOURCE CODE

FRONT END [HTML]

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Eco Mobile</title>
</head>
<body>
<h1>Welcome to Eco Mobile</h1>
<p>Your one-stop solution for mobile e-waste and scrap management!</p>
<p>Use the API to register, add items, and manage e-waste efficiently.</p>
</body>
</html>
```

BACK END

```
from flask import Flask, request, jsonify, render_template
```

```
app = Flask(__name__)
```

```
# Mock database users
```

```
= [] e_waste_items =
[]
```

```
# Homepage
```

```
@app.route("/") def
```

```
home():
```

```
return render_template("index.html")
```

```
# User Registration
```

```

@app.route("/register", methods=["POST"]) def
register_user():
    data = request.json user =
    {
        "username": data["username"], "email":
        data["email"],          "password":
        data["password"]
    } users.append(user)
    return jsonify({"message": "User registered successfully!", "users": users})

# Add E-Waste Item
@app.route("/add_item", methods=["POST"]) def
add_item():
    data = request.json item = {
        "item_name":      data["item_name"],
        "condition": data["condition"],
        "estimated_price": estimate_price(data["condition"])}
    e_waste_items.append(item)
    return jsonify({"message": "Item added successfully!", "item": item})

# Price Estimation Logic
def estimate_price(condition):
    condition_values = {"new": 80, "used": 50, "scrap": 20}
    return condition_values.get(condition.lower(), 0)

# List all E-Waste Items
@app.route("/list_items", methods=["GET"])
def list_items():
    return jsonify({"e_waste_items": e_waste_items})

if __name__ == "__main__":
    app.run(debug=True)

```

SAMPLE OUTPUT

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>EcoMobile</title>
</head>
<body>
  <h1>Welcome to EcoMobile!</h1>
  <p>Your platform for e-waste
management.</p>
</body>
</html>
```

Fig No: A1.1 Sample HTML Code

```
{
  "username": "john_doe",
  "email": "john@example.com",
  "password": "securepassword123"
}
```

```
{
  "item_name": "Smartphone",
  "condition": "Used"
}
```

```
{
  "message": "Item added
successfully!",
  "item": {
    "item_name": "Smartphone",
    "condition": "Used",
    "estimated_price": 50
  }
}
```

```
{
  "e_waste_items": [
    {
      "item_name":
"Smartphone",
      "condition": "Used",
      "estimated_price": 50
    }
  ]
}
```

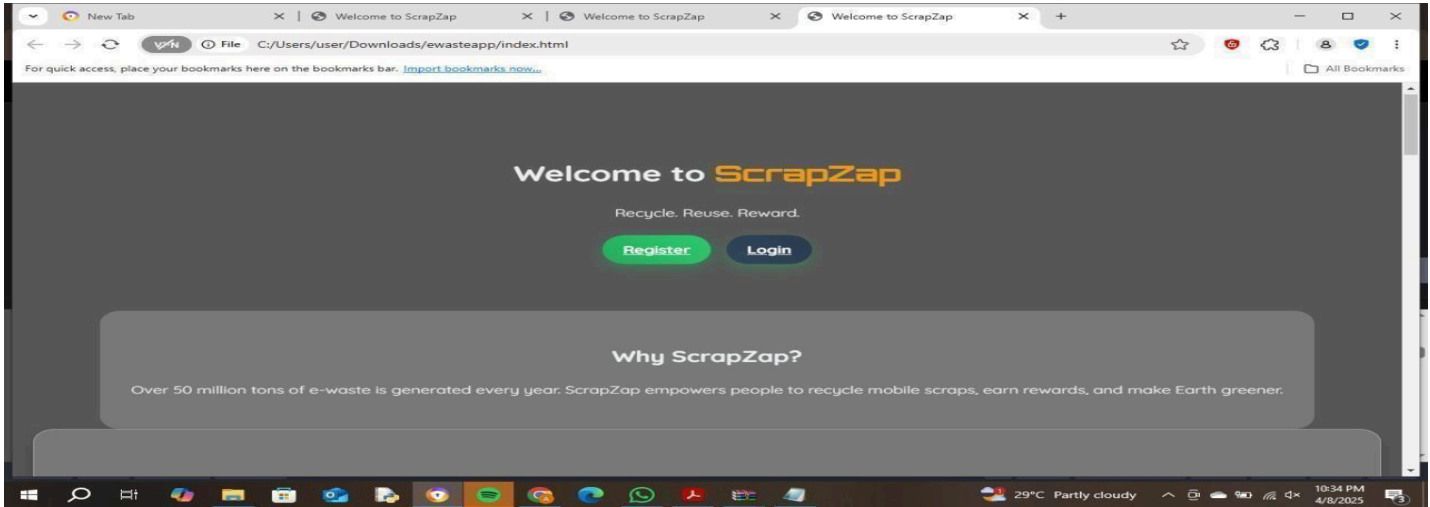
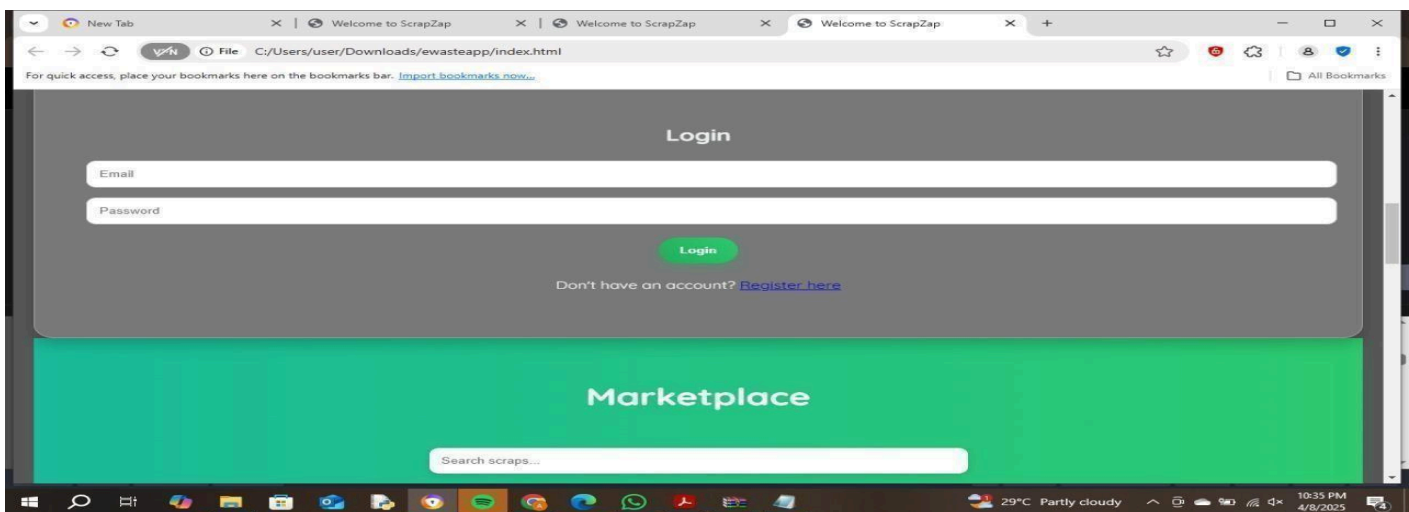
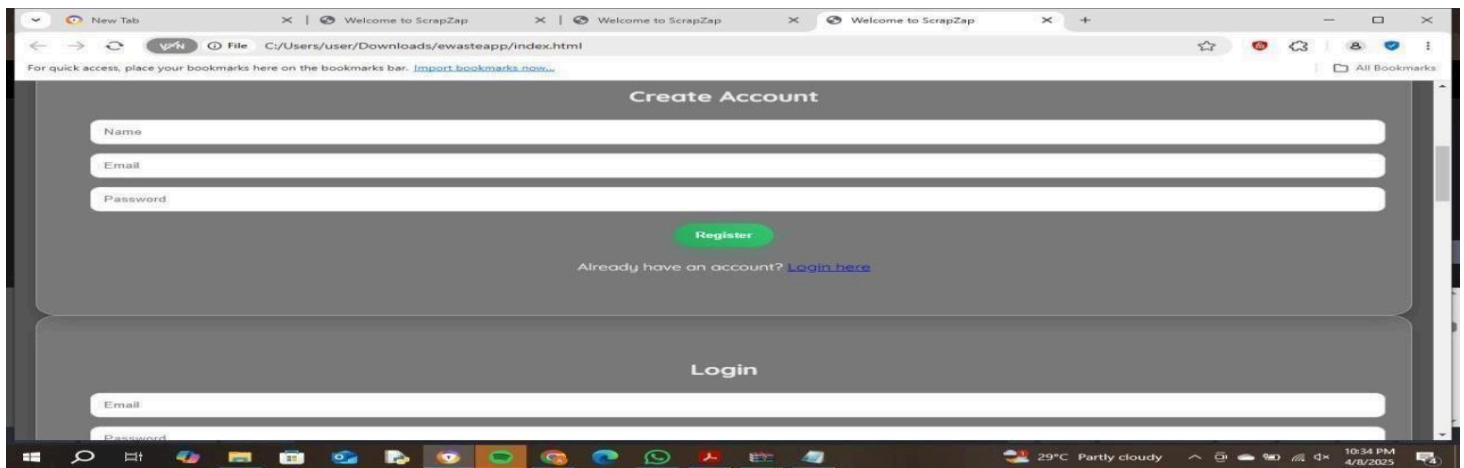
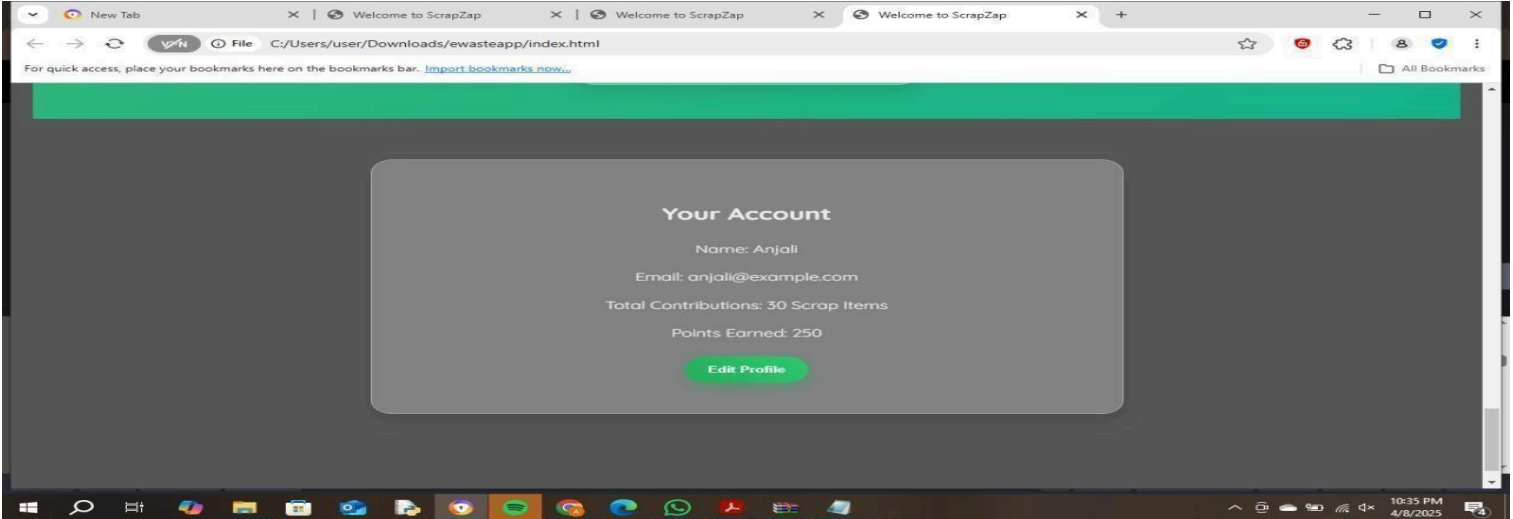
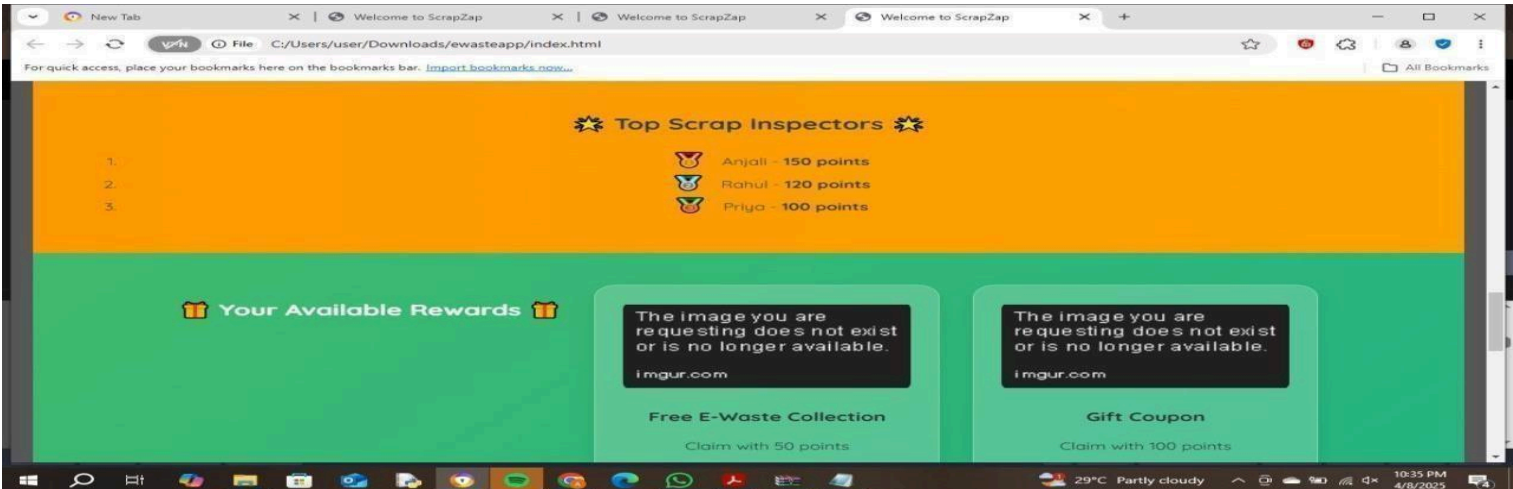
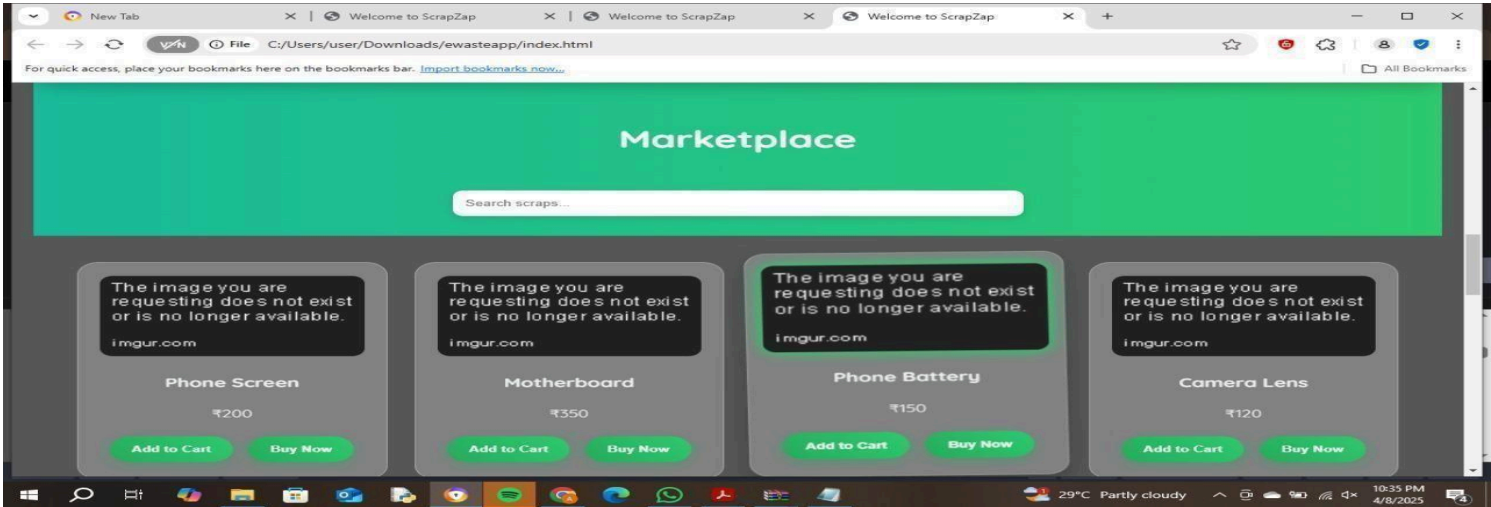


Fig No: A1.2 Login Page





APPENDIX II

KEY PERFORMANCE INDICATORS (KPI SECTION)

1. Participated in EMINENCE 1.0 conducted by Sairam Institutions in association with Start- Up India.



Fig No: A2.1 Event Participation

2. Achieved 1 st Prize under Goal No : 15 in the prestigious Solveathon competition.

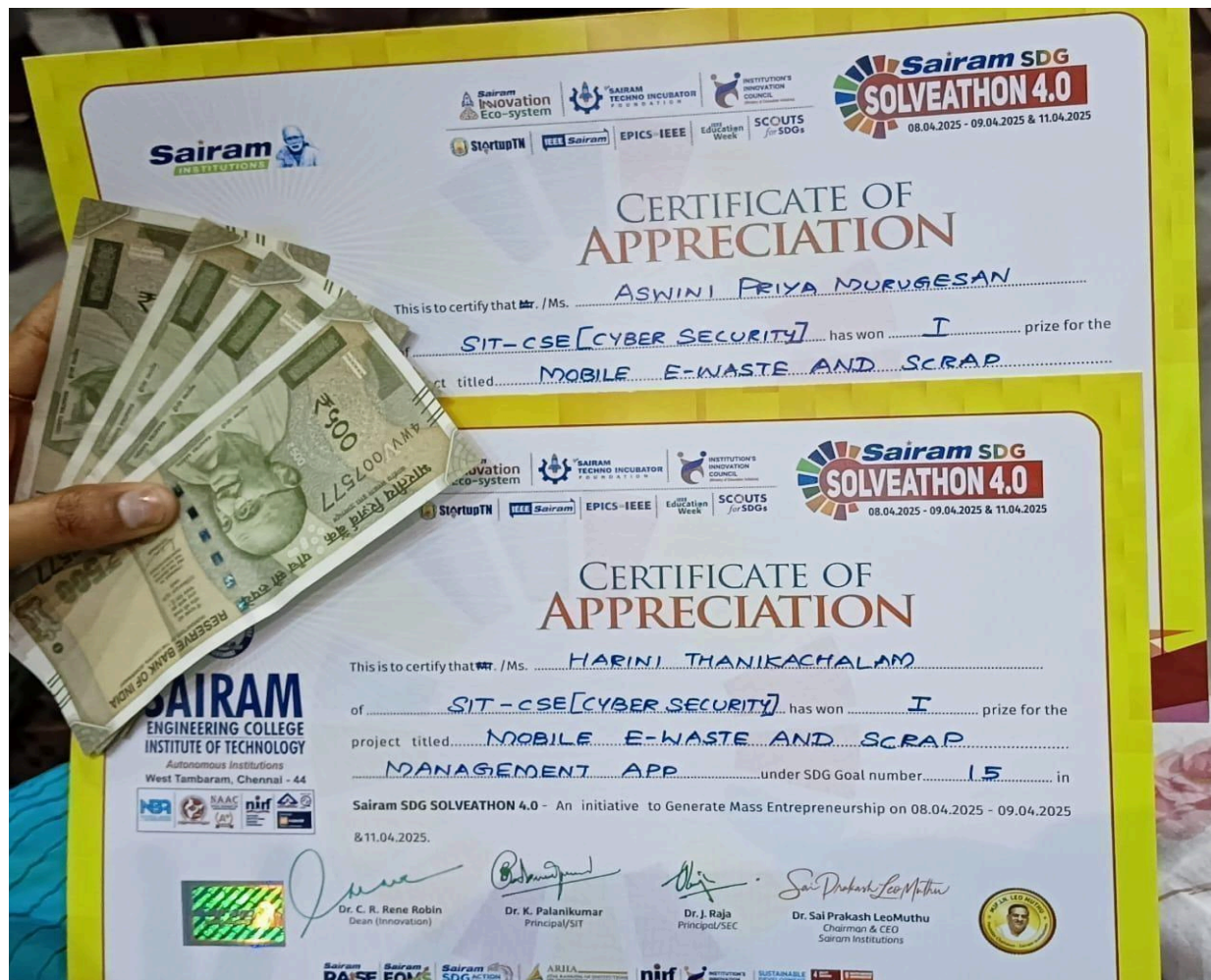


Fig No: A2.2 Solveathon Certificate

APPENDIX III

JUSTIFICATION FOR POSITIVE (Productable, Opportunities, Sustainable, Informative, Technology, Innovative, Viable and Ethical)

S.No	PARAMETERS	JUSTIFICATION	RATING (1 to 5)
1.	Productable	EcoReclaim	5
2.	Opportunities	Circularity	5
3.	Sustainable	EcoContinuity	5
4.	Informative	EcoInsight	4
5.	Technology	Eco Tech	4
6.	Innovative	EcoVate	5
7.	Viable	EcoSure	4
8.	Ethical	EcoEthos	4

Table A3.1 Justification for POSITIVE

Imagine the Future and Make it happen!



Together let's build a better world where there is **NO POVERTY** and **ZERO HUNGER**.
We have **GOOD HEALTH AND WELL BEING** **QUALITY EDUCATION** and full **GENDER EQUALITY** everywhere.

There is **CLEAN WATER AND SANITATION** for everyone. **AFFORDABLE AND CLEAN ENERGY**
which will help to create **DECENT WORK AND ECONOMIC GROWTH**. Our prosperity shall be fuelled
by investments in **INDUSTRY, INNOVATION AND INFRASTRUCTURE** that will help us to
REDUCE INEQUALITIES by all means. We will live in **SUSTAINABLE CITIES AND COMMUNITIES**.
RESPONSIBLE CONSUMPTION AND PRODUCTION will help in healing our planet.

CLIMATE ACTION will reduce global warming and we will have abundant,
flourishing **LIFE BELOW WATER**, rich and diverse **LIFE ON LAND**.

We will enjoy **PEACE AND JUSTICE** through **STRONG INSTITUTIONS**
and will build long term **PARTNERSHIPS FOR THE GOALS**.



For the goals to be reached,
everyone needs to do their part:
governments, the private sector,
civil society and **People like you.**

Together we can...

Sai Prakash Leo Mathur
Chairman & CEO - Sairam Institutions